



Confluence of Orthogonal DPRSs

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Outline

- 1. Introduction**
- 2. Deterministic Higher-Order Pattern Rewrite Systems**
- 3. Critical Pairs**
- 4. Orthogonal DPRSs**
- 5. Conclusion**

Definition (Miller 1991)

pattern is term in β -normal form such that arguments of free variables are η -equivalent to distinct bound variables

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Example

$$\begin{array}{l} x, y.F(y, x) \quad x.F(x, x) \quad x.F(c, x) \quad x.F(f(x), x) \quad x, y.F(f(x), f(y)) \\ x, y.F(x(y), x(c)) \quad x, y.G(y, z.x(z)) \quad x, y.G(x(y), z.x(z)) \quad x, y.G(y, z.x(z, c)) \end{array}$$

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Definition (Yokoyama & Hu & Takeichi 2003)

term in β -normal form is **deterministic higher-order pattern** (DHP) if for all subterms $\bar{x}_n.Y(\bar{s}_m)$

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unification problem for higher-order patterns is decidable

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Definitions

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Example (Yokoyama & Hu & Takeichi 2003)

- ▶ $f : a \rightarrow a$ $M, N : (a, a) \rightarrow a$
- ▶ terms $x, y.M(f(x), f(y))$ and $x, y.f(N(y, x))$ admit three (incomparable) unifiers:

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 - ① $\{M \mapsto z_1, z_2.z_1, N \mapsto z_1, z_2.z_2\}$

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- ▶ $f : a \rightarrow a$ $M, N : (a, a) \rightarrow a$ $Z : (a, a) \rightarrow a$
- ▶ terms $x, y.M(f(x), f(y))$ and $x, y.f(N(y, x))$ admit three (incomparable) unifiers:
 - 1 $\{M \mapsto z_1, z_2.z_1, N \mapsto z_1, z_2.z_2\}$
 - 2 $\{M \mapsto z_1, z_2.z_2, N \mapsto z_1, z_2.z_1\}$
 - 3 $\{M \mapsto z_1, z_2.f(Z(z_1, z_2)), N \mapsto z_1, z_2.Z(f(z_2), f(z_1))\}$

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Remarks

- ▶ we present **sound and complete unification procedure** for DHPs at IJCAR 2026

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Remarks

- ▶ we present sound and complete unification procedure for DHPs at IJCAR 2026
- ▶ minimal complete sets of unifiers may be infinite

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Example

- ▶ $\text{case} : (u, a \rightarrow i, b \rightarrow i) \rightarrow i$ $\text{inl} : a \rightarrow u$ $\text{inr} : b \rightarrow u$
- ▶ DPRS $\mathcal{R}$

$$\begin{aligned}\text{case}(\text{inl}(X), w.F(w), w.G(w)) &\rightarrow F(X) \\ \text{case}(\text{inr}(X), w.F(w), w.G(w)) &\rightarrow G(X) \\ \text{case}(Z, w.H(\text{inl}(w)), w.H(\text{inr}(w))) &\rightarrow H(Z)\end{aligned}$$

Definitions

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$\text{case}(Z, w.H(\text{inl}(w)), w.H(\text{inr}(w))) \rightarrow H(Z)$

- ▶ COPS #769
- ▶ 17 out of 124 HRSs in COPS are DPRSs

Example

► $\text{sum} (\text{foldr} (\lambda x, y. x^2 : ys) []) \rightsquigarrow \text{foldr} (\lambda x, y. x^2 + y) 0$

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$\text{sum}(\text{nil}) \rightarrow 0$

$\text{sum}(\text{cons}(x, xs)) \rightarrow x + \text{sum}(xs)$

$\text{comp}(z_3.\text{sum}(z_3), z_0.\text{foldr}_1(z_1, z_2.G(z_1, z_2), xs, z_0), ys) \rightarrow \text{check}(z_1, z_2.\text{sum}(G(z_1, z_2)), xs, ys)$

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► program transformation as rewriting:

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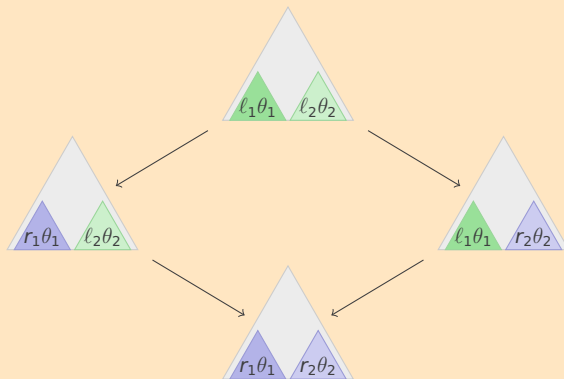
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$\rightarrow_{\mathcal{R}} z_0.\text{foldr}_2(z_1, z_2.z_1 \times z_1 + z_2, 0, z_0)$

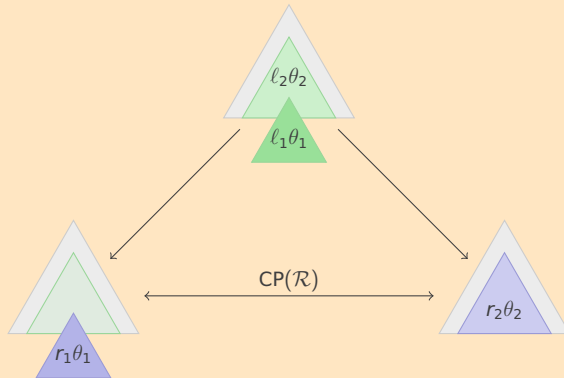
Outline

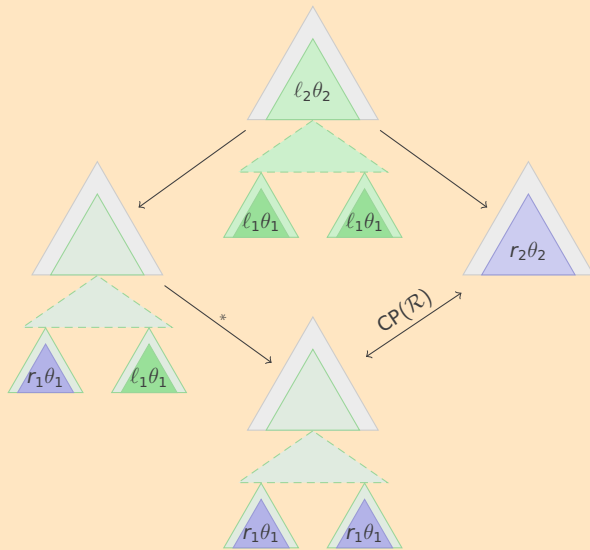
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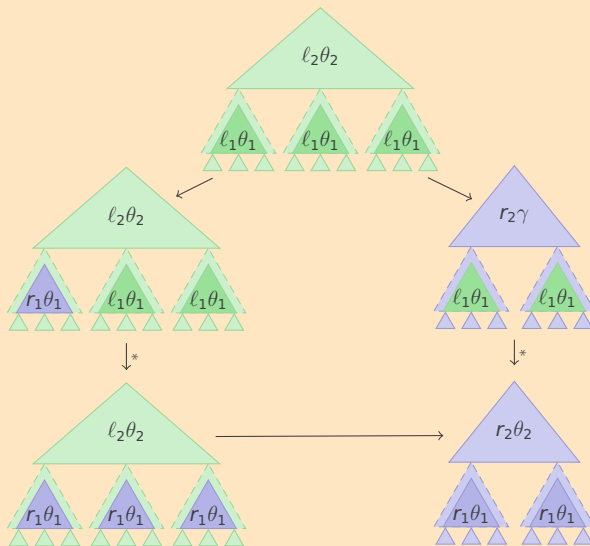
Parallel redexes



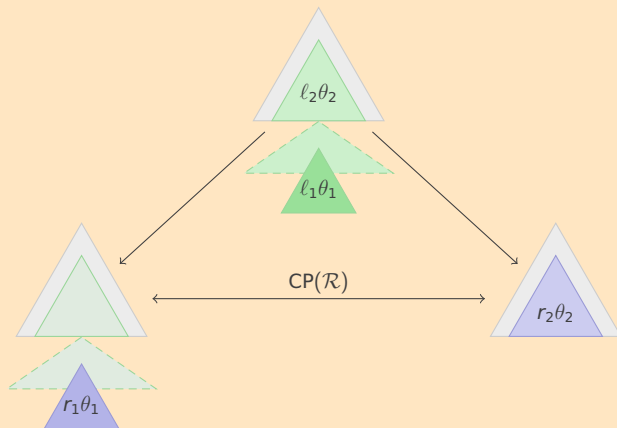
Standard Critical Overlap







Critical Variable Overlap



Theorem

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- ▶ more pressing issue: critical pairs often do not produce DHPs
- ▶ implementation available in prototype **hrstk**

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Definition

- ▶ DPRS $\mathcal{R}$ is **non-ambiguous** if $\text{CP}(\mathcal{R}) = \emptyset$

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- ▶ DPRS is **orthogonal** if it is both left-linear and non-ambiguous

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Definition

Given HRS $\mathcal{R}$, define multi-step relation inductively:

- ① $\bar{x}_k.h(\bar{s}_n) \multimap_{\mathcal{R}} \bar{x}_k.h(\bar{t}_n)$ if $\bar{x}_k.s_i \multimap_{\mathcal{R}} \bar{x}_k.t_i$ for all $1 \leq i \leq n$

Definition

- ▶ DPRS $\mathcal{R}$ is non-ambiguous if $\text{CP}(\mathcal{R}) = \emptyset$
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- ▶ generalized critical pairs to the rescue!

Lemma (Coherence)

for orthogonal DPRS $\mathcal{R}$, $\ell \rightarrow r \in \mathcal{R}$, $\ell \triangleright s$:

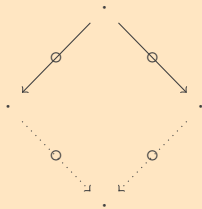
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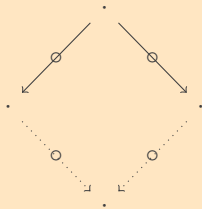


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Theorem

orthogonal DPRSs are confluent

Outline

1. Introduction
2. Deterministic Higher-Order Pattern Rewrite Systems
3. Critical Pairs
4. Orthogonal DPRSs
- 5. Conclusion**

Progress since IWC 2025

- ▶ sound and complete unification procedure

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